

The empirical recurrence for OEIS A316929, recovered from its tail

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Abstract

OEIS A316929 records “Empirical recurrence of order 89” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 112$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 93$. Its last coefficient is $c_{89} = 39936 \neq 0$, so no further tail satisfies anything shorter; any order-89 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316929 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 2, 3, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 82, 851, 20235, 467377, 10973407, 259460352, 6154985913, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Jul 16 2018 13:00:05, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 112$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 112$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 89: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 89.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 224$ exact terms from $n = 4$ on, it returns order 89 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	26	c_{24}	−4620954429867	c_{47}	26778136222531815	c_{70}	−445968697214742
c_2	−31	c_{25}	−11124133430809	c_{48}	48458424305825050	c_{71}	−30273255866210
c_3	59	c_{26}	−57872394788015	c_{49}	12888622919795952	c_{72}	159295500845201
c_4	−13088	c_{27}	−144357549815541	c_{50}	−23262575788370903	c_{73}	57503255636038
c_5	−24060	c_{28}	−48718536053893	c_{51}	−11741033426543995	c_{74}	−22954140803639
c_6	36526	c_{29}	546795691276052	c_{52}	5222480287053764	c_{75}	−20927991189291
c_7	1627305	c_{30}	1385787106076833	c_{53}	−10697150881254785	c_{76}	474280353200
c_8	6194923	c_{31}	1140751229677110	c_{54}	−33547754218799295	c_{77}	4243155195474
c_9	−661380	c_{32}	−1957021791551818	c_{55}	−12351045793199873	c_{78}	443192663886
c_{10}	−100600378	c_{33}	−6541363604097215	c_{56}	33382875798544555	c_{79}	−642703643380
c_{11}	−473077650	c_{34}	−6880904046334777	c_{57}	44794611803613026	c_{80}	−204983259092
c_{12}	−290036638	c_{35}	2991221998637586	c_{58}	12413620405690889	c_{81}	66058284728
c_{13}	3249312384	c_{36}	19000697586022756	c_{59}	−22806074844977019	c_{82}	36874838528
c_{14}	16067245663	c_{37}	22614399957553547	c_{60}	−24503136950558212	c_{83}	1035352944
c_{15}	19359337664	c_{38}	−453460225494303	c_{61}	−5315037877940470	c_{84}	−2277406880
c_{16}	−54569060541	c_{39}	−36872724652136236	c_{62}	8395708627007917	c_{85}	−483676480
c_{17}	−268681087834	c_{40}	−45068495490701088	c_{63}	6329165371591381	c_{86}	60825216
c_{18}	−418084763991	c_{41}	−3064175868459448	c_{64}	10911454147467	c_{87}	16084992
c_{19}	377419988212	c_{42}	53112281995414791	c_{65}	−2654136342020502	c_{88}	−1486336
c_{20}	2257319754006	c_{43}	55053067170896852	c_{66}	−596041205348821	c_{89}	39936
c_{21}	3420405129415	c_{44}	−6630005994802147	c_{67}	1154939760958361		
c_{22}	308980919537	c_{45}	−59962125077890863	c_{68}	735525307052833		
c_{23}	−5408571507820	c_{46}	−39331375287575751	c_{69}	−373722325868122		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 93$, and it is the recurrence OEIS A316929 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{89} - c_1x^{88} - \dots - c_{89}$. Its constant term is $-c_{89} = -39936$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 89 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 89, so $\deg q = 89$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 89, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 89, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 39936.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-89 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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